AttendSmart — Ultimate A-Z Team Workflow Guide

Complete 45-Day Development Program

⚠️ MANDATORY READING - NO EXCEPTIONS

This guide is your STRICT AI MENTOR that will ensure you complete AttendSmart in exactly 45 days. Follow every instruction precisely. No shortcuts, no skipping steps, no excuses.

GROUND RULES:
FAILURE IS NOT AN OPTION. FOLLOW THIS GUIDE EXACTLY.

✅ Daily check-ins mandatory - Report progress every evening
✅ Code commits required - Minimum 3 commits per day
✅ Testing before proceeding - Nothing moves forward without passing tests
✅ Documentation as you go - Write docs with every feature
✅ No feature creep - Stick to assigned tasks only
✅ Team coordination required - Daily standups and integration

TEAM ROLES & RESPONSIBILITIES

👨‍💻 KRISHNA - TECHNICAL LEAD (24%)

Your Sacred Mission: Build the Brain

- Face detection with smile liveness (Google ML Kit)
- BLE proximity detection and calibration
- Barometric sensor floor verification
- Encrypted database and security framework
- Backend API and real-time sync

👨‍💻 SHAMBHAVI - FLUTTER APP LEAD (20%)

Your Sacred Mission: Make It Beautiful & Functional

- Flutter app architecture and navigation
- UI screens and user interactions
- Sensor service integration
- State management with Provider/Riverpod
- Cross-platform compatibility

RAKSHITHA - UI/UX DESIGNER (13%)

Your Sacred Mission: Create Visual Magic

- Complete design system and style guide
- Icon creation and animations
- Accessibility compliance (contrast, scaling)
- Cinematic UI elements and motion
- Asset delivery and design handoff

ANANYA - DASHBOARD & ANALYTICS (13%)

Your Sacred Mission: Teacher's Command Center

- React dashboard with Material-UI
- Real-time analytics and charts
- QR code generation for large sessions
- Export functionality (CSV/PDF)
- Socket.io integration for live updates

HITHARTH - LOCATION & ADMIN (6%)

Your Sacred Mission: Smart Location Intelligence

- Geofencing and location services
- QR code backup functionality
- Notice management system
- Admin panel features
- Database optimization

YASHAS - KRISHNA'S SECURITY SHADOW (12%)

Your Sacred Mission: Quality & Security Guardian

- Security testing and audits
- BLE and sensor QA automation
- Performance optimization testing
- Bug detection and resolution
- Krishna's dedicated support system

NUHAN - INTEGRATION SPECIALIST (12%)

Your Sacred Mission: Make Everything Work Together

- Fingerprint UI and authentication
- Cross-platform testing
- API integration validation
- End-to-end testing automation
- UI/UX quality assurance

PHASE PRIORITY SYSTEM

CORE FEATURES (MUST COMPLETE - WEEKS 1-4)

Priority 1: Authentication & Security Foundation

- Student login (name, USN, DOB)
- Teacher authentication
- Encrypted local storage
- TLS 1.3 with certificate pinning

Priority 2: Biometric Verification Core

- Face detection with smile liveness
- BLE proximity scanning
- Barometric floor detection
- Multi-level verification system

Priority 3: Basic Attendance Flow

- Student check-in process
- Teacher session management
- Random peer verification
- Manual fallback system

Priority 4: Essential Dashboard

- Real-time attendance monitoring
- Basic analytics charts
- Session controls
- Export functionality

ENHANCED FEATURES (WEEKS 5-6)

- Advanced analytics
- Notification system
- QR code fallback
- Performance optimization
- Accessibility features

FUTURE FEATURES (POST-LAUNCH)

- Voice recognition
- Advanced biometrics
- Campus integration
- Parent notifications

ENVIRONMENT SETUP (DAY 0-1)

MANDATORY SOFTWARE INSTALLATION

Flutter Setup (Everyone)

```
# 1. Download Flutter 3.16.0 EXACT VERSION
# https://flutter.dev/docs/get-started/install
# 2. Extract to proper location
# Windows: C:\flutter
# Mac/Linux: /usr/local/flutter
export PATH="$PATH:/usr/local/flutter/bin"
# 3. Verify installation
flutter doctor -v
# 4. Fix ALL issues before proceeding
```

Android Studio Setup

```
# 1. Download from https://developer.android.com/studio
# 2. Install with ALL default options
# 3. SDK Manager -&gt; Install:
#    - Android 13 (API 33)
#    - Android 12 (API 32)
#    - Android 11 (API 30)
# 4. Create AVD: Pixel 6 Pro API 33
# 5. Name: AttendSmart_Test_Device
```

Required VS Code Extensions

1. Flutter (by Dart Code)
2. Dart (by Dart Code)

3. GitLens (by GitKraken)
4. Error Lens (by Alexander)
5. Bracket Pair Colorizer 2
6. Flutter Widget Snippets

REPOSITORY SETUP

Krishna Creates Repository (Day 0)

```
# Repository name: attendsmart-biometric-attendance
# Description: "Proxy-proof offline biometric attendance system"
# Initialize with README, .gitignore (Flutter)
git clone https://github.com/[username]/attendsmart-biometric-attendance.git
cd attendsmart-biometric-attendance
git checkout -b develop
git push origin develop
```

Team Members Clone (Day 1)

```
git clone https://github.com/[username]/attendsmart-biometric-attendance.git
cd attendsmart-biometric-attendance
git checkout develop
git pull origin develop
git checkout -b feature/[your-name]-setup
```

FOLDER STRUCTURE (FINAL)

```
/attendsmart-biometric-attendance
├── mobile/                          # Flutter App
│   ├── lib/
│   │   ├── core/                   # Krishna's Domain
│   │   │   ├── database/
│   │   │   ├── security/
│   │   │   └── constants/
│   │   ├── features/               # Shambhavi's Domain
│   │   │   ├── auth/
│   │   │   │   ├── data/
│   │   │   │   ├── domain/
│   │   │   │   └── presentation/
│   │   │   ├── attendance/
│   │   │   └── profile/
│   │   ├── shared/
│   │   │   ├── services/           # Krishna + Hitharth
│   │   │   │   ├── biometric/
│   │   │   │   ├── sensors/
│   │   │   │   └── location/
│   │   │   ├── widgets/            # Shambhavi + Rakshitha
│   │   │   └── utils/
│   │   ├── screens/                # Shambhavi's Domain
│   │   └── theme/                  # Rakshitha's Domain
```

```
│   ├── assets/                    # Rakshitha's Domain
│   │   ├── icons/
│   │   ├── images/
│   │   └── animations/
│   └── test/                      # Yashas + Nuhan
├── backend/                        # Krishna's Domain
│   ├── controllers/
│   ├── models/
│   ├── routes/
│   ├── middlewares/
│   └── services/
├── dashboard/                      # Ananya's Domain
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── services/
│   │   └── styles/
│   └── public/
├── docs/                          # Everyone Contributes
└── tests/                         # Yashas + Nuhan Lead
```

WEEKLY DETAILED WORKFLOW

WEEK 1: FOUNDATION (Days 1-7)

KRISHNA - Core Security & Database

Day 1: Database Architecture

- Create encrypted SQLite database
- Implement AES-256 encryption service
- Design tables: students, sessions, attendance
- Add comprehensive error handling
- **Commit**: "FEAT: Encrypted database foundation with security"

Day 2: Face Detection Service

- Integrate Google ML Kit Face Detection
- Implement smile liveness detection
- Add anti-spoofing measures
- Create confidence scoring system
- **Commit**: "FEAT: Face detection with smile liveness"

Day 3: BLE Proximity System

- Implement Flutter Blue Plus integration
- Create RSSI calibration system
- Add device discovery and filtering

- Implement range-based verification
- **Commit**: "FEAT: BLE proximity detection system"

Day 4: Barometric Floor Detection

- Integrate Sensors Plus barometer
- Implement floor calculation algorithms
- Add calibration and drift compensation
- Create floor verification service
- **Commit**: "FEAT: Barometric floor verification"

Day 5: Backend API Foundation

- Setup Node.js + Express server
- Create MongoDB connection and schemas
- Implement JWT authentication
- Add API rate limiting and security
- **Commit**: "FEAT: Secure backend API foundation"

Day 6: Multi-Level Verification

- Implement 4-tier verification system
- Create confidence fusion algorithms
- Add environmental multipliers
- Implement verification state machine
- **Commit**: "FEAT: Multi-level verification engine"

Day 7: Integration & Testing

- Test all services together
- Fix integration bugs
- Performance optimization
- Documentation updates
- **Commit**: "FIX: Core services integration and optimization"

SHAMBHAVI - Flutter App Foundation

Day 1: Project Architecture

- Create Flutter project structure
- Setup Provider for state management
- Implement GoRouter navigation
- Create base app configuration
- **Commit**: "FEAT: Flutter app architecture foundation"

Day 2: Authentication UI

- Design and implement login screens
- Create form validation
- Add loading states and error handling
- Implement secure storage integration
- **Commit**: "FEAT: Authentication UI with validation"

Day 3: Main Navigation

- Create bottom navigation bar
- Implement drawer navigation
- Add route guards and permissions
- Create navigation state management
- **Commit**: "FEAT: Complete app navigation system"

Day 4: Check-in Screen Foundation

- Design check-in flow UI
- Create camera preview screen
- Add sensor status indicators
- Implement progress indicators
- **Commit**: "FEAT: Check-in screen foundation"

Day 5: Service Integration

- Connect Krishna's biometric services
- Implement sensor data display
- Add real-time status updates
- Create error state handling
- **Commit**: "FEAT: Biometric service integration"

Day 6: Attendance History

- Create attendance history screen
- Implement data visualization
- Add filtering and search
- Create export functionality
- **Commit**: "FEAT: Attendance history and analytics"

Day 7: Testing & Polish

- Cross-platform testing
- UI/UX improvements
- Performance optimization
- Bug fixes and stability

- **Commit**: "FIX: UI polish and cross-platform compatibility"

RAKSHITHA - Design System

Day 1-3: Design Foundation

- Create complete color palette
- Design typography system
- Create icon library
- Setup design tokens

Day 4-5: Screen Designs

- Design all mobile screens
- Create accessibility guidelines
- Design dashboard layouts
- Create animation specifications

Day 6-7: Asset Delivery

- Export all design assets
- Create style guide documentation
- Implement theme configuration
- Test accessibility compliance

ANANYA - Dashboard Foundation

Day 1-3: React Setup

- Initialize React project
- Setup Material-UI theme
- Create basic layout structure
- Implement routing system

Day 4-5: Core Components

- Create attendance dashboard
- Implement real-time updates
- Add basic charts and analytics
- Create session management UI

Day 6-7: Backend Integration

- Connect to Krishna's API
- Implement WebSocket for real-time
- Add error handling
- Create export functionality

HITHARTH - Location Services

Day 1-3: Geofencing Setup

- Implement location permissions
- Setup geofencing service
- Test location accuracy
- Create campus boundary system

Day 4-5: QR Implementation

- Add QR code generation
- Implement QR scanning
- Create fallback mechanisms
- Test QR reliability

Day 6-7: Admin Features

- Create notice system
- Implement admin panels
- Add data management tools
- Test admin workflows

YASHAS - Security & QA

Day 1-7: Continuous Testing

- Write automated test scripts
- Perform security audits
- Test all Krishna's services
- Document security compliance
- Create performance benchmarks

NUHAN - Integration Testing

Day 1-7: E2E Testing

- Setup testing frameworks
- Create integration test suites
- Test UI components
- Validate API endpoints
- Document test procedures

WEEK 2: INTEGRATION (Days 8-14)

Focus: Connect All Components

- Complete service integration
- Real-time dashboard updates
- Peer verification system
- Manual fallback implementation

WEEK 3-4: FEATURE COMPLETION (Days 15-28)

Focus: Core Feature Polish

- Advanced analytics implementation
- Notification system
- Performance optimization
- Security hardening

WEEK 5-6: FINAL POLISH (Days 29-42)

Focus: Production Ready

- Comprehensive testing
- Accessibility compliance
- Performance tuning
- Deployment preparation

MANDATORY SECURITY REQUIREMENTS

Security Checklist (Never Skip)

✅ TLS 1.3 with certificate pinning
✅ AES-256 encryption for local data
✅ Device-bound JWT tokens
✅ Biometric data never leaves device
✅ GDPR/FERPA compliance
✅ Regular security audits

⚡ Performance Targets

✅ Check-in time: < 5 seconds
✅ Battery usage: < 40 mAh per session
✅ Face detection: < 2 seconds
✅ App launch: < 3 seconds

🔐 Verification Levels

Level 1: All sensors + high confidence → Instant approval
Level 2: 3/4 sensors pass → Auto approval with flag
Level 3: 2 sensors + peer verification → Teacher approval
Level 4: Manual entry only → Full audit trail

DAILY ACCOUNTABILITY

Mandatory Check-in Format (8 PM Daily)

```
📅 Day [X] Progress - [Your Name]

✅ COMPLETED:
- [Tasks finished with commit links]

⚠️ BLOCKED ON:
- [Current problems]
- [Help needed]

📋 TOMORROW:
- [Exact targets]

🧪 TESTS:
- [Tests run and results]

📸 PROOF:
- [Screenshots/videos]
```

GIT WORKFLOW RULES

Branch Strategy

- `main`: Production ready code only
- `develop`: Integration branch
- `feature/[name]-[task]`: Individual features
- `bugfix/[description]`: Bug fixes only

Commit Standards

```
# Good commits:
"FEAT: Add face detection with liveness check"
"FIX: Resolve BLE scanning timeout issue"
"TEST: Add unit tests for database operations"
"DOCS: Update API documentation"

# Bad commits (NOT ALLOWED):
"update code", "fix bug", "changes", "wip"
```

Pull Request Rules

- Clear title and description
- Link to related issues
- At least one reviewer approval
- All tests must pass
- No merge to main without full team review

TROUBLESHOOTING GUIDE

Emergency Procedures

```
# When stuck (try these first):
flutter clean
flutter pub get
# Restart IDE and computer if needed
```

Quality Gates

```
# Before any commit:
flutter format --set-exit-if-changed .
flutter analyze
flutter test
# Before any merge:
flutter build apk --debug
```

SUCCESS CRITERIA

Week 6 Deliverables

✓ Complete working app on Android and iOS
✓ All sensors working with >90% accuracy
✓ Real-time dashboard with analytics
✓ Comprehensive test suite >80% coverage
✓ Full documentation and user guides
✓ Security audit completion
✓ Performance benchmarks met
✓ Demo presentation ready

Non-Negotiable Requirements

- Privacy first: No biometric data leaves device
- Reliability: Works offline in any condition
- Security: End-to-end encryption standard
- Usability: Faster than manual attendance

FINAL MOTIVATION

Remember: Every role is CRITICAL to success. Krishna builds the intelligence, Shambhavi creates the experience, Rakshitha designs the beauty, Ananya provides insights, Hitharth adds intelligence, Yashas ensures quality, and Nuhan validates everything.

Together, you're building something that will transform education forever.

FOLLOW THIS GUIDE EXACTLY AND YOU WILL SUCCEED.

This guide is designed to be your ultimate AI mentor. Upload it to any AI system and it will guide you through every step of building AttendSmart successfully.